

Assignment: Structures in C Programming

1. Structure: Playlist

```
#include <stdio.h>

struct Playlist
{
    char title[50];
    char artist[50];
    int duration;
};

int main()
{
    struct Playlist song = {"Believer", "Imagine Dragons", 204};

    printf("Song Title: %s\n", song.title);
    printf("Artist: %s\n", song.artist);
    printf("Duration: %d seconds\n", song.duration);

    return 0;
}
```

Sample Output

Song Title: Believer

Artist: Imagine Dragons

Duration: 204 seconds

2. Structure: FoodItem Array

```
#include <stdio.h>
```

```
struct FoodItem
```

```
{
```

```
    char itemName[50];
```

```
    float price;
```

```
    float rating;
```

```
};
```

```
int main()
```

```
{
```

```
    struct FoodItem menu[3] =
```

```
    {
```

```
        {"Paneer Butter Masala", 249.0, 4.5},
```

```
        {"Veg Biryani", 199.0, 4.3},
```

```
        {"Masala Dosa", 129.0, 4.6}
```

```
    };
```

```
    printf("Zomato Menu Items:\n\n");
```

```
    for(int i = 0; i < 3; i++)
```

```
    {
```

```
        printf("Item: %s\n", menu[i].itemName);
```

```
        printf("Price: ₹%.2f\n", menu[i].price);
```

```
        printf("Rating: %.1f\n\n", menu[i].rating);
```

```
}  
  
    return 0;  
}
```

Sample Output

Item: Paneer Butter Masala

Price: ₹249.00

Rating: 4.5

Item: Veg Biryani

Price: ₹199.00

Rating: 4.3

Item: Masala Dosa

Price: ₹129.00

Rating: 4.6

3. Nested Structure: MovieShow

```
#include <stdio.h>
```

```
struct Time  
{  
    int hours;  
    int minutes;  
};
```

```
struct MovieShow
{
    char movie[50];
    int screen;
    struct Time showTime;
};

int main()
{
    struct MovieShow show =
    {
        "Avengers: Endgame",
        3,
        {7, 30}
    };

    printf("Movie: %s, Screen: %d, Time: %02d:%02d\n",
        show.movie,
        show.screen,
        show.showTime.hours,
        show.showTime.minutes);

    return 0;
}
```

Sample Output

Movie: Avengers: Endgame, Screen: 3, Time: 07:30

4. Nested Structure: InstaProfile

```
#include <stdio.h>
```

```
struct Bio
```

```
{  
    char description[100];  
    int age;  
};
```

```
struct InstaProfile
```

```
{  
    char username[50];  
    int followers;  
    struct Bio bio;  
};
```

```
int main()
```

```
{  
    struct InstaProfile profile =  
    {  
        "omprasad_nayak",  
        1500,  
        {"BCA Student | Tech Enthusiast", 19}  
    };  
}
```

```
printf("Username: %s\n", profile.username);  
printf("Followers: %d\n", profile.followers);  
printf("Bio: %s\n", profile.bio.description);  
printf("Age: %d\n", profile.bio.age);  
  
return 0;  
}
```

Sample Output

Username: omprasad_nayak

Followers: 1500

Bio: BCA Student | Tech Enthusiast

Age: 19

Conclusion

This assignment demonstrates:

- Creating and using structures in C.
- Initializing structure variables.
- Using arrays of structures.
- Working with nested structures.
- Accessing structure members using the dot (.) operator.

These concepts are useful for organizing and managing related data in C programs.